ParkChoiceSG Data Dictionary

Problem-domain terms aligned with the final ParkChoiceSG product and Lab 1 use cases

Term	Definition
Driver	A person using ParkChoiceSG to identify a suitable carpark for an intended destination. The driver does not need to create an account or profile.
Destination	The place the driver intends to visit. It is used to identify nearby candidate carparks and to determine the walking distance from each carpark.
Intended Arrival Time	The time at which the driver expects to arrive at a carpark. ParkChoiceSG uses this time when predicting how many parking lots are likely to be available upon arrival.
Parking Duration	The length of time the driver expects the vehicle to remain parked. It is used together with the applicable parking rates to estimate parking cost.
Carpark	A designated location where a driver may park a vehicle. A carpark may have information such as its name, location, type, height limit, parking rates and availability.
Candidate Carpark	A nearby carpark considered for a recommendation. A candidate is assessed using predicted availability at the intended arrival time, estimated parking cost, driving time and walking distance.
Selected Carpark	A carpark chosen by the driver from the recommendation results for further inspection or navigation.
Carpark Availability	The number of parking lots that are available at a carpark at a particular time. ParkChoiceSG distinguishes between a reported availability reading and the availability predicted for the driver's arrival time.
Current Reported Availability	The latest parking-lot availability reading obtained from an authorised parking data source. It represents availability at the reading time and is accumulated as part of the historical data used for prediction; it does not guarantee availability when the driver arrives.
Availability History	A collection of past availability readings for a carpark accumulated over time. The readings are grouped by factors such as carpark, day of week and time of day to support prediction of future availability.
Predicted Availability	The estimated number of parking lots expected to be available at a carpark at the driver's intended arrival time. It is derived from accumulated historical availability patterns rather than simply displaying the latest reported reading.
Prediction Confidence	An indication of the reliability of a predicted availability value, expressed as High, Medium or Low. It is determined from the quantity and variability of relevant historical availability observations.
Time-of-Day Bucket	A time interval used to group availability readings that occur at similar times of day for prediction purposes. The final bucket width may be configured by the project.
Day of Week	The day on which a parking observation or intended arrival occurs. It is used to match historical availability patterns for prediction.
Parking Rate	A published charging rule that determines the cost of parking during a specified day and time period. It may include free periods, grace periods and charge caps.
Estimated Parking Cost	The expected parking charge for a candidate carpark based on the intended arrival time, expected parking duration and applicable published parking rates. It is an estimate rather than a payment quotation or guarantee.
Driving Time	The estimated time required to travel to a candidate carpark under prevailing traffic conditions. It is one of the factors used to rank carparks.

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Walking Distance	The estimated distance from a candidate carpark to the driver's destination. It is one of the factors used to rank carparks.
Recommendation	A carpark suggestion produced after comparing suitable candidate carparks. ParkChoiceSG returns up to five recommendations ranked according to the Best Overall score.
Best Overall Score	A weighted score used to compare candidate carparks using four factors: predicted availability at the intended arrival time, estimated parking cost, driving time under live traffic and walking distance to the destination.
Recommendation Ranking	The ordering of suitable candidate carparks according to their Best Overall scores, with the highest-ranked carpark presented as the top recommendation.
Recommendation Justification	A concise explanation of why a carpark is recommended based on the factors contributing to its ranking. The top recommendation includes a one-line justification.
Data Freshness	The extent to which retrieved carpark or routing information is recent enough to be relied upon. Information that is stale or unavailable is identified to the driver rather than presented as current.
Data Staleness Indicator	A visible indication that displayed information is older than the desired freshness level, particularly when ParkChoiceSG must rely on cached data because an external data source is unavailable.
Carpark Information	The information presented for a selected carpark, such as its name, location, predicted availability and confidence, estimated parking cost, driving time, walking distance, relevant restrictions and recommendation explanation.
Carpark Status	The operational state of a carpark, such as open or closed. A closed carpark should not be treated as an eligible recommendation.
Saved Location	A location, such as Home or Work, saved by the driver for convenient reuse. Saved locations belong to the driver and are retained only on the driver's device.
Favourite Carpark	A carpark marked by the driver for convenient future access. Favourite carparks are retained only on the driver's device.
Search History	A record of the driver's previous destination searches retained locally on the driver's device for convenient reuse.
Parking Session	The period during which the driver's vehicle is parked at a selected carpark. A session may track its start time, expected duration, accruing estimated cost and expected expiry.
Parking Reminder	A local reminder associated with a parking session that alerts the driver when the expected parking duration is approaching its end.
Anonymous Feedback	A report submitted about incorrect, outdated or problematic carpark information without requiring the driver to create an account or identify themselves.
External Navigation Service	A separate navigation service that receives the selected carpark as the destination and provides routing or turn-by-turn navigation. ParkChoiceSG performs the handoff but does not provide turn-by-turn navigation itself.
Destination Coordinates	The geographic coordinates of a destination or selected carpark used to identify its location and support mapping, distance calculation and navigation handoff.
Operator	An authorised administrator who reviews anonymous reports, may correct or override carpark information such as closure status, and monitors the health of external data integrations.